GCaMP model figure reproduction files

This repository contains the data and plotting code used to reproduce the figures, supplementary figures, and parameter tables for the manuscript submission. Most figure panels are generated by MATLAB scripts inside the corresponding figure* or supp_figure* directory; Python-based panels are exposed through setup.py console entry points.

Environment

Use MATLAB for .m scripts and a conda/venv Python environment for .py scripts. From the repository root:

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -e .
```

Conda works equivalently after activating the target environment. The Python entry points resolve data and output paths back to this project checkout, not to the installed environment location. If launching from an unusual location, set GCAMP_FIGSHARE_ROOT or FIGSHARE_PROJECT_DIR to the repository root.

MATLAB scripts should be run from the repository root or from their panel directory after adding the repository and Utilities directory to the MATLAB path. Scripts write outputs to local output folders when an output path is part of the script.

Main figures

- figure1: figure1_B_C_G_I_code.m, figure1_D_E_F_code.m, figure1_H_J_code.m, figure1_K_L_M_code.m, figure1_N_code.m, figure1_O_code.m.
- figure2: figure2_C_D_code.m, figure2_E_F_code.m, figure2_G_H_I_code.m, figure2_J_code.m, figure2_K_L_code.m.
- figure3: MATLAB panels figure3_B_code.m, figure3_D_E_code.m; Python panels via run_figure3_F_G_code.
- figure4: MATLAB panels figure4B_C_code.m, figure4_D_code.m; Python panel via run_figure4_E_code. Current Figure 4 correlation data are in figure4/correlation_tbl_jG8f.mat.

Supplementary figures and tables

- supp_figure1 to supp_figure5: MATLAB scripts in each directory generate the corresponding panels.
- supp_figure6: suppfigure_6_code.m and make_gcamp_params_table.py generate the parameter summary material; rendered table outputs are included.
- supp_figure9 to supp_figure11: updated numbering reflects the inserted supplementary figure. supp_figure11 Python panel runs with run_supp_figure11_A_code.
- supp_figure12: run_supp_figure12_code generates the trialwise raincloud and trace-panel outputs from packaged CSV/NPZ exports.
- supp_figure13: suppfigure13_A_B_code.m uses janelia_model_selection_data_jG8f.mat; run_supp_figure13_C_D_code generates the trialwise parameter comparison panels.
- supp_figure14: suppfigure14_A_B_C_code.m uses decay_v_spikeNum.mat for quiescent binning panels, suppfigure14_D_E_code.m uses janelia_model_selection_data_jG8m.mat, and run_supp_figure14_F_G_code generates the jG8m trialwise comparison panels.
- Table2 and Table3: CSV, TeX, PDF, and render scripts for manuscript parameter tables.

Shared utilities and data

Shared_data contains common MATLAB data sources. Utilities contains shared MATLAB helpers, including AccessFitFiles_Janelia.m, GCaMPalias.m, and color utilities. SharedUtilities contains shared Python helpers, including trialwise_plots.py and path-resolution helpers used by installed console scripts.

Intermediate model-selection files in supp_figure13 and supp_figure14 intentionally contain only the fit information needed for model selection and plotting; the raw fit directories are not required by the panel scripts.

Outputs

Generated figures are written to the relevant figure directory or its output subdirectory. Python scripts accept --output-dir, --format, and --show where applicable. MATLAB panel scripts generally use local defaults matching the packaged data. Temporary cache directories created by the supplementary Python scripts are cleaned at the end of each run.